INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

OFFICIAL ACADEMIC LEARNING RESOURCE â¢ DEPARTMENT OF CSE

Course: [BCS-303] Digital Electronics

Programme: Bachelor of Technology in Computer Science & Engineering **Resource Type: LECTURE NOTES**

Module / Unit: Unit 1 **Faculty Author: Prof. R. K. Patel (Dept. of CSE**

Unit 1 Lecture Notes: Number Systems, Boolean Logic & K-Maps

==========

1. Radix Conversions & Signed Arithmetic

- Base Conversions: Binary, Octal, Hexadecimal radix mathematical representations.
- Signed Binary Formats: Sign-magnitude, 1’s complement, and 2’s complement arithmetic.
- Binary Codes: BCD (8421), Excess-3, Gray Code (unit distance code), ASCII encoding.

2. Boolean Algebra Axioms & Logic Minimization

- Postulates: Commutative, Associative, Distributive, Identity, and De Morgan’s Laws.
- Canonical Forms: Sum of Products (SOP / minterms) and Product of Sums (POS / maxterms).
- Karnaugh Maps (K-Maps): 2, 3, and 4-variable grouping, prime implicants, and don’t care conditions.